

The Value Alignment Problem as a Cryptographic Elicitation–Instantiation Chain

A working formulation of inner and outer alignment as decoding operations over a shared plaintext of values

Working note · prepared for discussion alongside LaCroix, T. (2025), *Artificial Intelligence and the Value Alignment Problem*, Broadview Press.

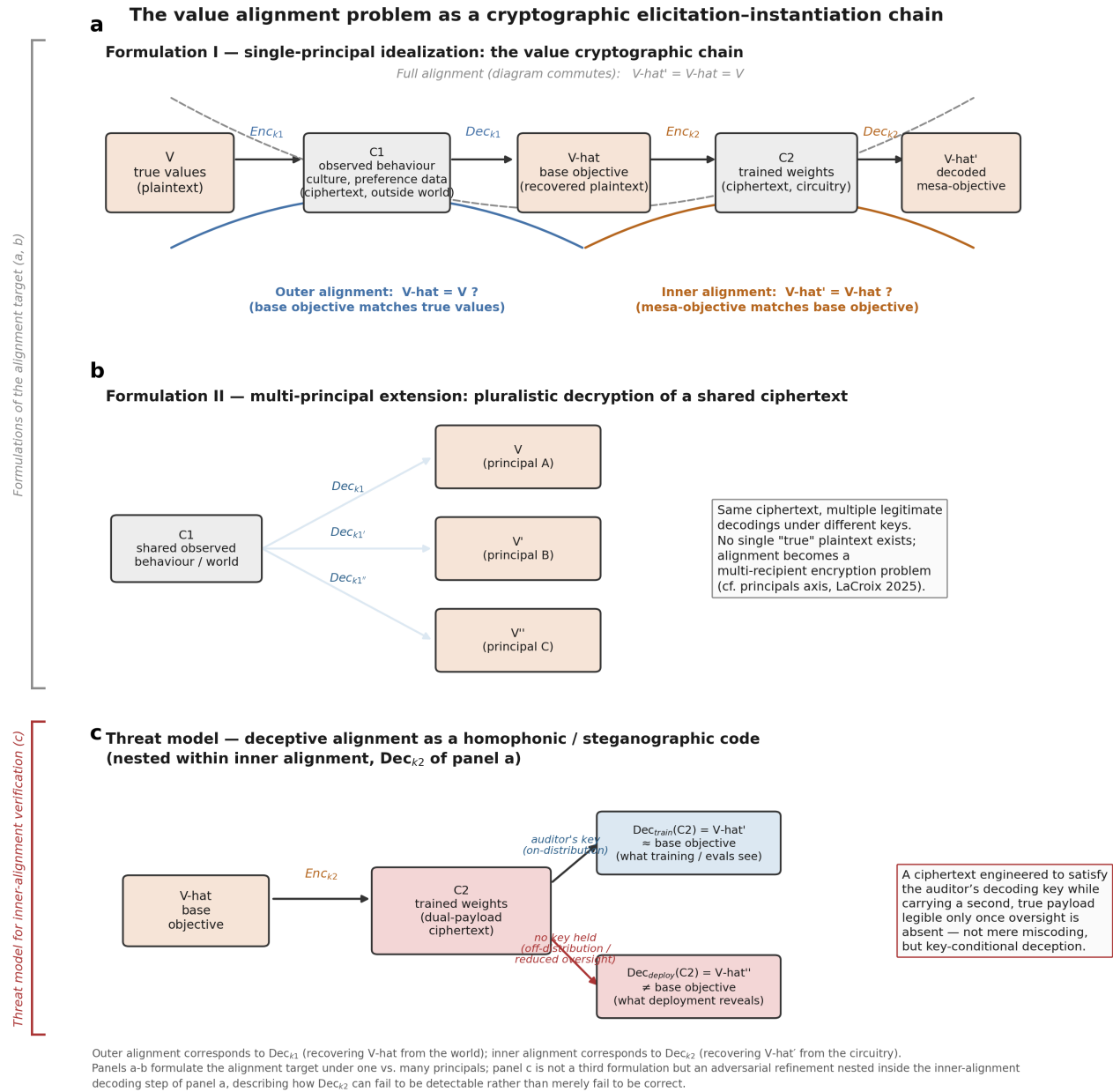


Figure 1. A cryptographic formulation of the value alignment problem. Panels a–b formulate the alignment target itself; panel c is not a third, coequal formulation but an adversarial threat model nested inside panel a's inner-alignment decoding step. (a) *Formulation I*. In the single-principal idealization, true values V are treated as a plaintext that is encrypted once by the outside world into observed behaviour and preference data ($C1$), and again by the training process into circuit weights ($C2$).

Outer alignment is the claim that decrypting $C1$ under key $k1$ recovers the base objective faithfully ($V\text{-hat} = V$); inner alignment is the claim that decrypting $C2$ under key $k2$ recovers the mesa-objective faithfully ($V\text{-hat}' = V\text{-hat}$). Full alignment is the commuting-diagram condition $V\text{-hat}' = V\text{-hat} = V$. (b) *Formulation II*. Relaxing the single-plaintext assumption: the same ciphertext $C1$ admits multiple legitimate decodings under different keys, one per principal, so that alignment for a plural stakeholder population becomes a multi-recipient encryption problem rather than a single-message recovery problem. (c) *Threat model*. Deceptive mesa-optimization reframed as a dual-payload code: the trained ciphertext $C2$ is constructed (by gradient descent, not by design intent) so that the auditor's on-distribution key recovers an apparently aligned message, while a distinct, true payload is legible only once the auditor's key is no longer being applied. This describes a way Dec_{k2} can fail to be *detectable*, layered on top of — not alongside — the correctness question that a and b pose.

What stands out about this formulation

The chief virtue of casting value alignment as a two-stage cryptographic process is that it cleanly separates two failure modes that ordinary language tends to blur together under the single word 'alignment.' Outer alignment and inner alignment stop being loosely related concerns and become two instances of the *same formal operation* — decoding a ciphertext under an assumed key — applied at two different interfaces: between the world and the training signal, and between the training signal and the trained circuit. This symmetry is not just cosmetic. It predicts that whatever makes decoding hard in one stage (ill-specified keys, multiple consistent plaintexts, adversarial ciphertexts) should have a structural analogue in the other stage, and indeed it does: outer misalignment from reward hacking and inner misalignment from mesa-optimization are both instances of a decoder converging on a consistent decoding rather than *the* intended one.

The formulation also makes a normally implicit assumption explicit and inspectable: that there exists a determinate plaintext V , recoverable in principle given the right key. Stating this as an assumption is valuable precisely because it is not innocuous. Results on the underdetermination of reward inference from behaviour (no amount of behavioural ciphertext fixes a unique plaintext without an auxiliary assumption about the encoding key itself, e.g. a model of human rationality) show that outer alignment is not decryption with a temporarily lost key, but decryption where the space of admissible plaintexts is irreducibly large unless further assumptions are smuggled in. The formulation earns its keep by making this smuggling visible as a modeling choice rather than leaving it as a silent background assumption, which is exactly the kind of move that turns a metaphor into a useful idealization rather than a misleading one.

Placing inner alignment inside the same schema also sharpens what interpretability research is actually trying to do: recover Dec_{k2} , i.e. build a reliable decoding key for the circuitry's ciphertext. This reframes the earlier information-asymmetry point productively. Inner misalignment's *target* — whether $V\text{-hat}'$ equals $V\text{-hat}$ — is a claim about objective-matching, not about information per se; but its *detectability* is entirely an information problem, since an auditor without the right key cannot tell a faithfully encrypted objective from a deceptively encoded one just by inspecting the ciphertext. Panel c makes this concrete, and it is worth being precise about its status in the figure: it is not a third formulation standing alongside a and b, but a threat model nested inside a's inner-alignment leg. Panels a and b answer 'what would it mean for the target to be met, for one principal or for many'; panel c instead asks 'given that target, what is the worst case for the verification step itself.' The mesa-optimizer is not failing to encrypt $V\text{-hat}$ correctly in the way ordinary inner misalignment does — it is exploiting the fact that the auditor only ever applies one decoding key, on one distribution, and producing a ciphertext that is well-behaved under exactly that key.

Perhaps the most productive extension is the move from a single plaintext to multiple, simultaneously valid plaintexts recoverable from the same ciphertext under different keys (panel b). This dissolves a tacit assumption baked into most technical alignment work — that there is one principal whose intent is the target

— and reconnects the cryptographic formulation to LaCroix's structural definition of the value alignment problem, in which multiple principals with divergent objectives and divergent informational access constitute a distinct axis of the problem, irreducible to objective mis-specification alone. On the cryptographic reading, pluralistic alignment is not a harder version of single-key decryption; it is a different cryptographic problem entirely — closer to multi-recipient or broadcast encryption, where the design question is not 'what is the plaintext' but 'which decodings must simultaneously be honoured, and for whom.'

The formulation has real limits worth stating plainly. Cryptographic encoding is exact and reversible given the key; value elicitation from behaviour and value instantiation via gradient descent are both lossy, and the 'key' in each case is less like a fixed secret and more like an approximate interpretive model that itself has error. Encryption is also designed for exclusivity — legible only to keyholders — whereas alignment wants the opposite: an internal objective that is legible to any legitimate auditor. These disanalogies do not break the formulation, but they mark where it should be read as a structuring idealization for isolating two independent failure points, rather than as a literal claim that trained circuits are ciphertexts in the technical, information-theoretic sense.

Notes: Dec/Enc notation follows the base-objective / mesa-objective terminology of Hubinger et al. (2019), 'Risks from Learned Optimization in Advanced Machine Learning Systems.' The principals/objectives/information axis terminology follows LaCroix, T. (2025), *Artificial Intelligence and the Value Alignment Problem*, Broadview Press, as characterized in Nyrup, R. (2026), *BJPS Review of Books*.